

### LECTURE 3

#### SIMULTANEOUS EQUATIONS II: MULTIPLE-EQUATION GMM, 3SLS

In this lecture, we consider joint GMM estimation of more than one simultaneous equation. As we will see, joint estimation can lead to efficiency gains.

#### Multiple-equation GMM estimator

Suppose that all  $m$  equations in (1) of Lecture 2 are identified. Using the notation from Lecture 2, equation (7), we can write the system of  $m$  equations as follows:

$$\begin{aligned} y_{1i} &= X_{1,i}^\top \delta_1 + u_{1i}, \\ &\dots \\ y_{mi} &= X_{m,i}^\top \delta_m + u_{mi}, \end{aligned}$$

where for all  $j = 1, \dots, m$ ,  $\delta_j \in \mathbb{R}^{k_j}$ ,  $k_j = m_j + l_j$ , and the random  $l$ -vector  $Z_i$  is such that

$$\begin{aligned} \text{rank}(\mathbb{E}[Z_i X_{j,i}^\top]) &= k_j, \\ \mathbb{E}[Z_i u_{ji}] &= 0. \end{aligned}$$

Equivalently, the system can be rewritten in matrix notation as

$$\begin{aligned} y_1 &= X_1 \delta_1 + u_1, \\ &\dots \\ y_m &= X_m \delta_m + u_m, \end{aligned}$$

where, for  $j = 1, \dots, m$ ,  $X_j$  collects the  $n$  observations on the right-hand side variables in the  $j$ -th equation:

$$X_j = \begin{pmatrix} X_{j,1}^\top \\ \vdots \\ X_{j,n}^\top \end{pmatrix},$$

$y_j$  collects the  $n$  observations on the left-hand side variable in the  $j$ -th equation:

$$y_j = \begin{pmatrix} y_{j1} \\ \vdots \\ y_{jn} \end{pmatrix},$$

and  $u_j$  is defined similarly. Further, define

$$\begin{aligned} X &= \begin{pmatrix} X_1 & 0 \\ & \dots \\ 0 & X_m \end{pmatrix}, \\ Y &= \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, \\ U &= \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix}, \end{aligned}$$

$$\delta = \begin{pmatrix} \delta_1 \\ \vdots \\ \delta_m \end{pmatrix}.$$

The matrix  $X$  is  $(nm) \times k$ , where  $k = k_1 + \dots + k_m$ ,  $Y$  and  $U$  are  $(nm) \times 1$ , and  $\delta$  is  $k \times 1$ . The system can now be compactly written as

$$Y = X\delta + U.$$

In this system, we have  $ml$  population moment conditions:

$$E \begin{bmatrix} Z_i u_{1i} \\ \vdots \\ Z_i u_{mi} \end{bmatrix} = E[U_i \otimes Z_i] = 0,$$

where  $U_i = (u_{1i}, \dots, u_{mi})^\top$ , and  $A \otimes B$  denotes the Kronecker product of  $A$  and  $B$ .<sup>1</sup> To define sample moment conditions that can be used for estimation, consider the  $ml$ -vector of sample covariations between the exogenous variables and errors:

$$\begin{pmatrix} Z^\top u_1 \\ \vdots \\ Z^\top u_m \end{pmatrix} = \begin{pmatrix} Z^\top (y_1 - X_1 \delta_1) \\ \vdots \\ Z^\top (y_m - X_m \delta_m) \end{pmatrix},$$

where  $Z$  denotes the  $n \times l$  matrix of observations on the exogenous variables. Using the Kronecker product notation, this can be conveniently written as:

$$(I_m \otimes Z)^\top U = (I_m \otimes Z)^\top (Y - X\delta).$$

Let  $A_n$  be an  $(ml) \times (ml)$  weight matrix. The system or multiple-equation GMM estimator is obtained by solving

$$\min_{d \in \mathbb{R}^k} (Y - Xd)^\top (I_m \otimes Z) A_n^\top A_n (I_m \otimes Z)^\top (Y - Xd).$$

Thus, the system GMM estimator is given by<sup>2</sup>

$$\hat{\delta} = (X^\top (I_m \otimes Z) A_n^\top A_n (I_m \otimes Z)^\top X)^{-1} X^\top (I_m \otimes Z) A_n^\top A_n (I_m \otimes Z)^\top Y.$$

Define

$$W_n = A_n^\top A_n,$$

and introduce a partition

$$W_n = \begin{pmatrix} W_{11,n} & \dots & W_{1m,n} \\ \dots & \dots & \dots \\ W_{m1,n} & \dots & W_{mm,n} \end{pmatrix},$$

where each  $W_{ij,n}$  is an  $l \times l$  matrix (with  $W_{ij,n} = W_{ji,n}^\top$ ). The system GMM estimators for the  $m$  equations can be written as

$$\begin{pmatrix} \hat{\delta}_1 \\ \vdots \\ \hat{\delta}_m \end{pmatrix} =$$

---

<sup>1</sup>Suppose that  $A$  is  $k \times l$  and  $B$  is  $m \times n$ . Then  $A \otimes B$  is a  $(km) \times (ln)$  matrix given by

$$A \otimes B = \begin{pmatrix} a_{11} & \dots & a_{1l} \\ \dots & \dots & \dots \\ a_{k1} & \dots & a_{kl} \end{pmatrix} \otimes B = \begin{pmatrix} a_{11}B & \dots & a_{1l}B \\ \dots & \dots & \dots \\ a_{k1}B & \dots & a_{kl}B \end{pmatrix}.$$

The properties of the Kronecker product include:  $(A \otimes B)^\top = A^\top \otimes B^\top$ ,  $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$  when  $A$  and  $B$  are invertible, and  $(A \otimes B)(C \otimes D) = AC \otimes BD$  for properly defined matrices  $C$  and  $D$ .

<sup>2</sup>Recall that in the single equation case, the GMM estimator solves  $\min_b (Y - Xb)^\top Z A_n^\top A_n Z^\top (Y - Xb)$ , and the solution is given by  $\hat{\beta} = (X^\top Z A_n^\top A_n Z^\top X)^{-1} X^\top Z A_n^\top A_n Z^\top Y$ .

$$\begin{aligned}
&= \left( \begin{pmatrix} X_1^\top Z & 0 \\ 0 & X_m^\top Z \end{pmatrix} \begin{pmatrix} W_{11,n} & \dots & W_{1m,n} \\ \dots & \dots & \dots \\ W_{m1,n} & \dots & W_{mm,n} \end{pmatrix} \begin{pmatrix} Z^\top X_1 & 0 \\ 0 & Z^\top X_m \end{pmatrix} \right)^{-1} \\
&\quad \times \begin{pmatrix} X_1^\top Z & 0 \\ 0 & X_m^\top Z \end{pmatrix} \begin{pmatrix} W_{11,n} & \dots & W_{1m,n} \\ \dots & \dots & \dots \\ W_{m1,n} & \dots & W_{mm,n} \end{pmatrix} \begin{pmatrix} Z^\top y_1 \\ \vdots \\ Z^\top y_m \end{pmatrix} \quad (1) \\
&= \begin{pmatrix} X_1^\top Z W_{11,n} Z^\top X_1 & \dots & X_1^\top Z W_{1m,n} Z^\top X_m \\ \dots & \dots & \dots \\ X_m^\top Z W_{m1,n} Z^\top X_1 & \dots & X_m^\top Z W_{mm,n} Z^\top X_m \end{pmatrix}^{-1} \begin{pmatrix} X_1^\top Z W_{11,n} Z^\top y_1 + \dots + X_1^\top Z W_{1m,n} Z^\top y_m \\ \dots \\ X_m^\top Z W_{m1,n} Z^\top y_1 + \dots + X_m^\top Z W_{mm,n} Z^\top y_m \end{pmatrix}. \quad (2)
\end{aligned}$$

We can compare the above expression with that for equation-by-equation GMM:

$$\begin{pmatrix} \tilde{\delta}_1 \\ \vdots \\ \tilde{\delta}_m \end{pmatrix} = \begin{pmatrix} X_1^\top Z A_{1n}^\top A_{1n} Z^\top X_1 & 0 \\ 0 & X_m^\top Z A_{mn}^\top A_{mn} Z^\top X_m \end{pmatrix}^{-1} \begin{pmatrix} X_1^\top Z A_{1n}^\top A_{1n} Z^\top y_1 \\ \dots \\ X_m^\top Z A_{mn}^\top A_{mn} Z^\top y_m \end{pmatrix}.$$

From the comparison, it is apparent that the equation-by-equation GMM estimator is a particular case of the system GMM estimator with weighting matrices  $W_{ij,n} = 0$  for  $i \neq j$ .

## Large-sample properties of the multiple-equation GMM estimator

From (1), we can write

$$\begin{aligned}
&\begin{pmatrix} \hat{\delta}_1 - \delta_1 \\ \vdots \\ \hat{\delta}_m - \delta_m \end{pmatrix} = \\
&= \left( \begin{pmatrix} \sum_{i=1}^n X_{1,i} Z_i^\top & 0 \\ 0 & \sum_{i=1}^n X_{m,i} Z_i^\top \end{pmatrix} \begin{pmatrix} W_{11,n} & \dots & W_{1m,n} \\ \dots & \dots & \dots \\ W_{m1,n} & \dots & W_{mm,n} \end{pmatrix} \begin{pmatrix} \sum_{i=1}^n Z_i X_{1,i}^\top & 0 \\ 0 & \sum_{i=1}^n Z_i X_{m,i}^\top \end{pmatrix} \right)^{-1} \\
&\quad \times \begin{pmatrix} \sum_{i=1}^n X_{1,i} Z_i^\top & 0 \\ 0 & \sum_{i=1}^n X_{m,i} Z_i^\top \end{pmatrix} \begin{pmatrix} W_{11,n} & \dots & W_{1m,n} \\ \dots & \dots & \dots \\ W_{m1,n} & \dots & W_{mm,n} \end{pmatrix} \begin{pmatrix} \sum_{i=1}^n Z_i u_{1i} \\ \vdots \\ \sum_{i=1}^n Z_i u_{mi} \end{pmatrix}.
\end{aligned}$$

In addition to the previous assumptions, we assume:

- $\{(Y_i^\top, Z_i^\top) : i \geq 1\}$  are i.i.d..
- $W_n \rightarrow_p W$ , where  $W$  is positive definite (and symmetric).
- The elements of  $U_i = (u_{1i}, \dots, u_{mi})^\top$  and  $Z_i$  have finite second moments (which together with the reduced form equations implies that  $E[Z_i X_{j,i}^\top]$  is finite for all  $j = 1, \dots, m$ ).

Under these assumptions we have consistency of the system GMM estimator:  $\hat{\delta}_j \rightarrow_p \delta_j$  for all  $j = 1, \dots, m$ . Next, for asymptotic normality we also assume that:

- The elements of  $U_i$  and  $Z_i$  have finite fourth moments (which implies that the elements of  $Y_i$  have finite fourth moments as well).

Under these assumptions,

$$n^{-1/2} \begin{pmatrix} \sum_{i=1}^n Z_i u_{1i} \\ \vdots \\ \sum_{i=1}^n Z_i u_{mi} \end{pmatrix} \rightarrow_d N(0, \Omega),$$

where

$$\begin{aligned} \Omega &= E \left[ \begin{pmatrix} Z_i u_{1i} \\ \vdots \\ Z_i u_{mi} \end{pmatrix} \begin{pmatrix} Z_i u_{1i} \\ \vdots \\ Z_i u_{mi} \end{pmatrix}^\top \right] \\ &= E \left[ (U_i \otimes Z_i) (U_i \otimes Z_i)^\top \right] \\ &= E \left[ U_i U_i^\top \otimes Z_i Z_i^\top \right]. \end{aligned}$$

Then, we have that

$$n^{1/2} \begin{pmatrix} \hat{\delta}_1 - \delta_1 \\ \vdots \\ \hat{\delta}_m - \delta_m \end{pmatrix} \rightarrow_d N(0, V(W)),$$

where

$$\begin{aligned} V(W) &= (C^\top W C)^{-1} C^\top W \Omega W C (C^\top W C)^{-1}, \\ C &= \begin{pmatrix} Q_1 & 0 \\ 0 & Q_m \end{pmatrix}, \\ Q_j &= E[Z_i X_{j,i}^\top], \text{ for } j = 1, \dots, m. \end{aligned}$$

Let us assume that

- $\Omega$  is positive definite.

As usual, the efficient GMM estimator corresponds to  $W_n$  that satisfies

$$W_n \rightarrow_p \Omega^{-1}.$$

For example,

$$\begin{aligned} W_n &= \hat{\Omega}_n^{-1} \\ &= \left( n^{-1} \sum_{i=1}^n \left( \hat{U}_i \hat{U}_i^\top \otimes Z_i Z_i^\top \right) \right)^{-1}, \end{aligned}$$

where  $\hat{U}_i$  is constructed using some preliminary consistent estimators of the  $\delta_j$ 's, for example, equation-by-equation 2SLS estimators:

$$\hat{U}_i = \begin{pmatrix} y_{1i} - X_{1,i}^\top \tilde{\delta}_1^{2SLS} \\ \vdots \\ y_{mi} - X_{m,i}^\top \tilde{\delta}_m^{2SLS} \end{pmatrix}.$$

The asymptotic variance of the efficient GMM estimator is given by

$$V(\Omega^{-1}) = (C^\top \Omega^{-1} C)^{-1}. \quad (3)$$

## Homoskedastic errors

In the case of homoskedastic errors, i.e., if

$$\mathbb{E} [U_i U_i^\top | Z_i] = \Sigma,$$

where  $\Sigma$  is some positive definite  $m \times m$  matrix, we have

$$\begin{aligned} \Omega &= \mathbb{E} [U_i U_i^\top \otimes Z_i Z_i^\top] \\ &= \mathbb{E} [\mathbb{E} [U_i U_i^\top | Z_i] \otimes Z_i Z_i^\top] \\ &= \Sigma \otimes \mathbb{E} [Z_i Z_i^\top]. \end{aligned}$$

In this case,

$$\hat{\Omega}_n = \hat{\Sigma}_n \otimes n^{-1} \sum_{i=1}^n Z_i Z_i^\top,$$

where

$$\hat{\Sigma}_n = n^{-1} \sum_{i=1}^n \hat{U}_i \hat{U}_i^\top.$$

A GMM estimator with

$$W_n = \hat{\Sigma}_n^{-1} \otimes \left( n^{-1} \sum_{i=1}^n Z_i Z_i^\top \right)^{-1}$$

is called the three-stage LS estimator (3SLS).

The 3SLS estimator can be written in the matrix notation as follows:

$$\begin{aligned} \hat{\delta}^{3SLS} &= \left( X^\top (I_m \otimes Z) \left( \hat{\Sigma}_n^{-1} \otimes (Z^\top Z)^{-1} \right) (I_m \otimes Z)^\top X \right)^{-1} X^\top (I_m \otimes Z) \left( \hat{\Sigma}_n^{-1} \otimes (Z^\top Z)^{-1} \right) (I_m \otimes Z)^\top Y \\ &= \left( X^\top \left( \hat{\Sigma}_n^{-1} \otimes \left( Z (Z^\top Z)^{-1} Z^\top \right) \right) X \right)^{-1} X^\top \left( \hat{\Sigma}_n^{-1} \otimes \left( Z (Z^\top Z)^{-1} Z^\top \right) \right) Y \\ &= \left( X^\top \left( \hat{\Sigma}_n^{-1} \otimes P_Z \right) X \right)^{-1} X^\top \left( \hat{\Sigma}_n^{-1} \otimes P_Z \right) Y, \end{aligned}$$

where  $P_Z = Z(Z^\top Z)^{-1}Z^\top$ . Further, let  $\hat{\sigma}^{ij}$  be the  $(i, j)$ -th element of  $\hat{\Sigma}_n^{-1}$ , and define

$$\begin{aligned} \hat{X}_j &= P_Z X_j, \\ \hat{X} &= \begin{pmatrix} \hat{X}_1 & 0 \\ \vdots & \vdots \\ 0 & \hat{X}_m \end{pmatrix}. \end{aligned}$$

We have:

$$\begin{aligned} \begin{pmatrix} \hat{\delta}_1^{3SLS} \\ \vdots \\ \hat{\delta}_m^{3SLS} \end{pmatrix} &= \begin{pmatrix} \hat{\sigma}^{11} X_1^\top P_Z X_1 & \dots & \hat{\sigma}^{1m} X_1^\top P_Z X_m \\ \vdots & \ddots & \vdots \\ \hat{\sigma}^{m1} X_m^\top P_Z X_1 & \dots & \hat{\sigma}^{mm} X_m^\top P_Z X_m \end{pmatrix}^{-1} \\ &\quad \times \begin{pmatrix} \hat{\sigma}^{11} X_1^\top P_Z y_1 + \dots + \hat{\sigma}^{1m} X_1^\top P_Z y_m \\ \vdots \\ \hat{\sigma}^{m1} X_m^\top P_Z y_1 + \dots + \hat{\sigma}^{mm} X_m^\top P_Z y_m \end{pmatrix} \\ &= \begin{pmatrix} \hat{\sigma}^{11} \hat{X}_1^\top X_1 & \dots & \hat{\sigma}^{1m} \hat{X}_1^\top X_m \\ \vdots & \ddots & \vdots \\ \hat{\sigma}^{m1} \hat{X}_m^\top X_1 & \dots & \hat{\sigma}^{mm} \hat{X}_m^\top X_m \end{pmatrix}^{-1} \begin{pmatrix} \hat{\sigma}^{11} \hat{X}_1^\top y_1 + \dots + \hat{\sigma}^{1m} \hat{X}_1^\top y_m \\ \vdots \\ \hat{\sigma}^{m1} \hat{X}_m^\top y_1 + \dots + \hat{\sigma}^{mm} \hat{X}_m^\top y_m \end{pmatrix} \end{aligned}$$

$$= \left( \hat{X}^\top \left( \hat{\Sigma}_n^{-1} \otimes I_n \right) X \right)^{-1} \hat{X}^\top \left( \hat{\Sigma}_n^{-1} \otimes I_n \right) Y.$$

Using (3), the asymptotic variance of the 3SLS estimator is given by the inverse of

$$\begin{aligned} & \begin{pmatrix} Q_1 & 0 \\ 0 & Q_m \end{pmatrix}^\top \left( \Sigma^{-1} \otimes (E[Z_i Z_i^\top])^{-1} \right) \begin{pmatrix} Q_1 & 0 \\ 0 & Q_m \end{pmatrix} \\ &= \begin{pmatrix} Q_1 & 0 \\ 0 & Q_m \end{pmatrix}^\top \begin{pmatrix} \sigma^{11} (E[Z_i Z_i^\top])^{-1} & \dots \sigma^{1m} (E[Z_i Z_i^\top])^{-1} \\ \dots & \dots \sigma^{mm} (E[Z_i Z_i^\top])^{-1} \end{pmatrix} \begin{pmatrix} Q_1 & 0 \\ 0 & Q_m \end{pmatrix}, \end{aligned}$$

where

$$\Sigma^{-1} = \begin{pmatrix} \sigma^{11} & \dots & \sigma^{1m} \\ \dots & \dots & \dots \\ \sigma^{m1} & \dots & \sigma^{mm} \end{pmatrix}.$$

Thus, the asymptotic variance of the 3SLS estimator is

$$V(\Omega^{-1}) = \begin{pmatrix} \sigma^{11} Q_1^\top (E[Z_i Z_i^\top])^{-1} Q_1 & \dots & \sigma^{1m} Q_1^\top (E[Z_i Z_i^\top])^{-1} Q_m \\ \dots & \dots & \dots \\ \sigma^{m1} Q_m^\top (E[Z_i Z_i^\top])^{-1} Q_1 & \dots & \sigma^{mm} Q_m^\top (E[Z_i Z_i^\top])^{-1} Q_m \end{pmatrix}^{-1}.$$

## Single- vs. multiple-equation GMM

As we have seen, the equation-by-equation GMM corresponds to the case where  $W_n$  has a diagonal structure. Since in general  $\Omega^{-1}$  does not take this form, even the “efficient” version of the equation-by-equation GMM estimator is inefficient when compared to the multiple-equation GMM. The reason for this is that the single equation estimator for equation  $j$  ignores the information about that equation contained in other equations. However, there are two exceptions to that rule.

First, let us assume conditional homoskedasticity. Suppose further that the errors are uncorrelated across the equations, i.e.,  $\Sigma$  is diagonal:

$$\begin{aligned} \Sigma &= \begin{pmatrix} E[u_{1i}^2] & \dots E[u_{1i} u_{mi}] \\ \dots & \dots \\ E[u_{mi} u_{1i}] & \dots E[u_{mi}^2] \end{pmatrix} \\ &= \begin{pmatrix} E[u_{1i}^2] & \dots 0 \\ \dots & \dots \\ 0 & \dots E[u_{mi}^2] \end{pmatrix} \\ &= \begin{pmatrix} \sigma_1^2 & \dots 0 \\ \dots & \dots \\ 0 & \dots \sigma_m^2 \end{pmatrix}. \end{aligned}$$

In this case,

$$\begin{aligned} & V(\Omega^{-1}) \\ &= \left( \begin{pmatrix} Q_1^\top & 0 \\ 0 & Q_m^\top \end{pmatrix} \begin{pmatrix} \sigma_1^2 E[Z_i Z_i^\top] & 0 \\ 0 & \sigma_m^2 E[Z_i Z_i^\top] \end{pmatrix}^{-1} \begin{pmatrix} Q_1 & 0 \\ 0 & Q_m \end{pmatrix} \right)^{-1} \\ &= \begin{pmatrix} \sigma_1^2 (Q_1^\top (E[Z_i Z_i^\top])^{-1} Q_1)^{-1} & 0 \\ 0 & \sigma_m^2 (Q_m^\top (E[Z_i Z_i^\top])^{-1} Q_m)^{-1} \end{pmatrix}. \end{aligned}$$

First, the estimators are asymptotically independent across the equations. Second, the asymptotic variance of the multiple-equation efficient GMM estimator is the same as that of the equation-by-equation 2SLS estimator. Hence, in the homoskedastic case and when the errors are uncorrelated across the equations (conditional on the  $Z_i$ 's), the equation-by-equation 2SLS estimator is efficient.

The equation-by-equation estimator is also efficient when all equations are exactly identified. In fact, single- and multiple-equation estimators are the same when the system is exactly identified:

$$\begin{aligned}
\begin{pmatrix} \hat{\delta}_1 \\ \vdots \\ \hat{\delta}_m \end{pmatrix} &= \left( X^\top (I_m \otimes Z) W_n (I_m \otimes Z)^\top X \right)^{-1} X^\top (I_m \otimes Z) W_n (I_m \otimes Z)^\top Y \\
&= \left( (I_m \otimes Z)^\top X \right)^{-1} W_n^{-1} (X^\top (I_m \otimes Z))^{-1} X^\top (I_m \otimes Z) W_n (I_m \otimes Z)^\top Y \\
&= \left( (I_m \otimes Z)^\top X \right)^{-1} (I_m \otimes Z)^\top Y \\
&= \begin{pmatrix} (Z^\top X_1)^{-1} Z^\top y_1 \\ \vdots \\ (Z^\top X_m)^{-1} Z^\top y_m \end{pmatrix}.
\end{aligned}$$

Thus, when the system is exactly identified, all the estimators discussed so far reduce to the equation-by-equation IV estimator.